3.2 Factoring Special Products

Name _____ Period ____ Date _____

Algebra II 3.2 · Student sheet

I can…

- factor the difference of two squares.
- factor perfect square trinomials.
- factor completely and solve equations using special product patterns.

Vocabulary · fill in the missing word

Difference of two squares — $a^2 - b^2 =$ _____

Perfect square trinomial — $a^2 + 2ab + b^2 =$ _____ and $a^2 - 2ab + b^2 = (a - b)^2$

Factored completely — no factor can be factored _____

Part 1 · Mental math challenge

Use a pattern (no calculator!). Show the pattern you used.

Product	Pattern	Answer
$31 \cdot 29$		
$102 \cdot 98$		
$45 \cdot 55$		
21^2		

Part 2 · Notes: fill in the blanks

$a^2 - b^2 =$ _____

$a^2 + 2ab + b^2 =$ _____ $a^2 - 2ab + b^2 =$ _____

A sum of two squares, $a^2 + b^2$, _____.

Part 3 · Examples (work them with the class)

Example 1

Factor each expression.

a. $x^2 - 49$ b. $9y^2 - 16$

Example 2

Factor each expression.

a. $x^2 - 10x + 25$ b. $4x^2 + 12x + 9$

Example 3

Factor completely: $3x^3 - 12x$

Example 4

Solve each equation.

a. $x^2 - 14x + 49 = 0$ b. $4x^2 = 25$

Part 4 · Partner practice

Partner 1

Factor $25x^2 - 1$.

Partner 2

Factor completely: $2x^2 + 8x + 8$